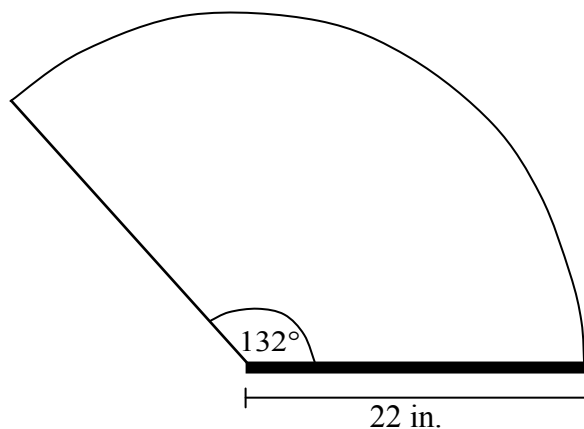


Question 1

T1

A section of a car windshield is cleaned by a wiper as shown in the diagram below. The arm of the wiper is 22 inches, and it rotates through a central angle of 132° . Determine the length of the arc that is created by the tip of the wiper.



Solution

$$\theta = 132 \times \frac{\pi}{180}$$

1 mark for conversion

$$= \frac{132\pi}{180} \text{ or } \frac{11\pi}{15}$$

$$s = \theta r$$

$$s = \frac{11\pi}{15} (22)$$

1 mark for substitution

$$s = \frac{242\pi}{15} \text{ inches}$$

2 marks

or

$$s = 50.684 \text{ inches}$$

There are 20 boys and 11 girls who can be selected to be on a team.

Determine the number of ways that 7 boys and 5 girls can be selected for this team.

Solution

$${}_{20}C_7 \cdot {}_{11}C_5 = 35\,814\,240$$

½ mark for ${}_{20}C_7$

½ mark for ${}_{11}C_5$

1 mark for the product of combinations

2 marks

Question 3

R10

A water filtration system which removes impurities from a sample of water can be modelled by

$$P = 0.25(0.55)^n, \text{ where:}$$

P = the percentage of impurities remaining, in decimal form

n = the number of filters

Determine, algebraically, how many filters are required so that less than 1% of the impurities remain in the water sample. Express your answer as a whole number.

Solution

$$0.01 = 0.25(0.55)^n \quad \frac{1}{2} \text{ mark for substitution}$$

$$0.04 = (0.55)^n$$

$$\log(0.04) = n \log(0.55) \quad \frac{1}{2} \text{ mark for applying logarithms}$$

$$\frac{\log(0.04)}{\log(0.55)} = n \quad \frac{1}{2} \text{ mark for power law}$$

$$n = 5.384 \ 203 \quad \frac{1}{2} \text{ mark for solving for } n$$

$\therefore$ 6 filters are needed

2 marks

In the binomial expansion of $\left(x^2 - \frac{2}{y}\right)^8$, determine the middle term in simplified form.

Solution

$$t_5 = {}_8C_4 \left(x^2\right)^{8-4} \left(-\frac{2}{y}\right)^4 \quad 2 \text{ marks (1 mark for } {}_8C_4, \frac{1}{2} \text{ mark for each consistent factor)}$$

$$t_5 = 70x^8 \left(\frac{16}{y^4}\right)$$

$$t_5 = \frac{1120x^8}{y^4}$$

1 mark for simplification ($\frac{1}{2}$ mark for coefficient, $\frac{1}{2}$ mark for exponents)

3 marks

Solve the following equation algebraically over the interval $[0, 2\pi]$.

$$6\sin^2 \theta + \sin \theta - 1 = 0$$

Solution

$$(3\sin \theta - 1)(2\sin \theta + 1) = 0$$

$$3\sin \theta - 1 = 0$$

$$2\sin \theta + 1 = 0$$

$$\sin \theta = \frac{1}{3}$$

$$\sin \theta = -\frac{1}{2}$$

1 mark for solving for $\sin \theta$ ($\frac{1}{2}$ mark for each branch)

$$\theta_r = 0.339836$$

$$\theta = 0.340$$

$$\theta = \frac{7\pi}{6}$$

$$\theta = 2.802$$

$$\theta = \frac{11\pi}{6}$$

2 marks ($\frac{1}{2}$ mark for each value of θ)

or

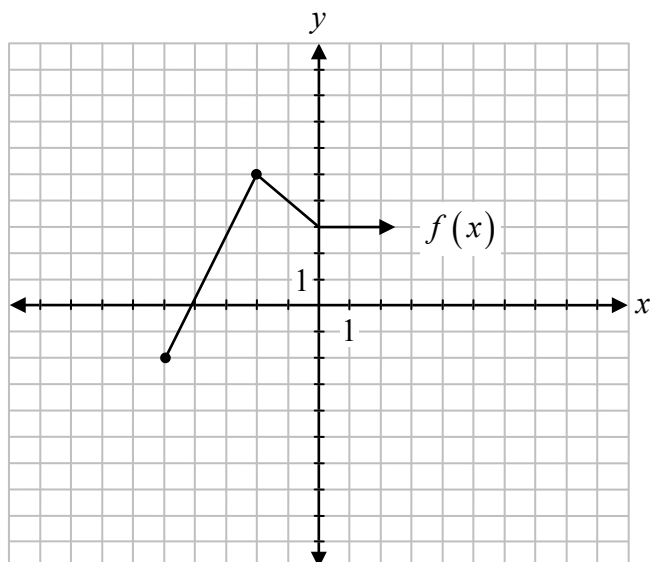
$$\theta = 0.340, 2.802, 3.665, 5.760$$

3 marks

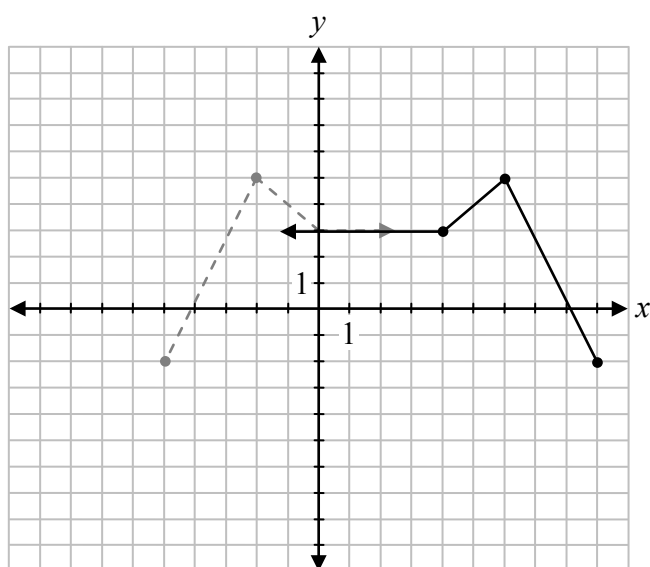
Question 6

R4, R5

Given the graph of $y = f(x)$, sketch the graph of $y = f(-x + 4)$.



Solution



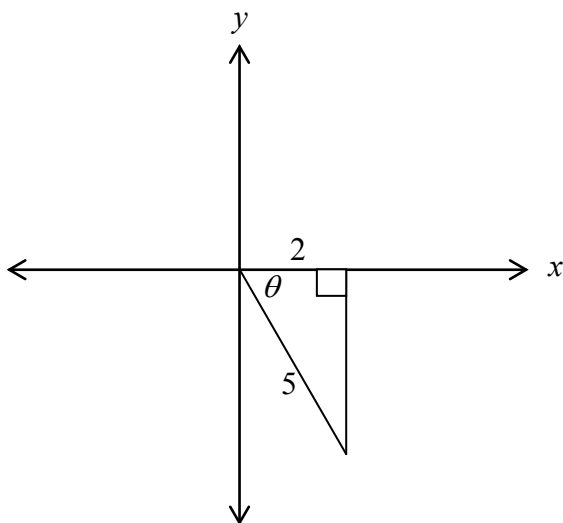
1 mark for reflection over the y -axis
1 mark for horizontal translation

2 marks

Question 7

T2, T3

Given the following triangle, determine $\csc \theta$.



Solution

$$x^2 + y^2 = r^2$$

$$2^2 + y^2 = 5^2$$

½ mark for substitution

$$y^2 = 21$$

$$y = \pm\sqrt{21}$$

½ mark for solving for y

$$\csc \theta = -\frac{5}{\sqrt{21}}$$

1 mark for $\csc \theta$ (½ mark for quadrant, ½ mark for value)

2 marks

Note(s):

- Accept any of the following values for y : $y = \pm\sqrt{21}$, $y = \sqrt{21}$, or $y = -\sqrt{21}$.

Solve algebraically.

$${}_nP_2 = 9n$$

Solution

$$\frac{n!}{(n-2)!} = 9n$$

$$\frac{n(n-1)\cancel{(n-2)!}}{\cancel{(n-2)!}} = 9n$$

$$n(n-1) = 9n$$

$$n^2 - n = 9n$$

$$n^2 - 10n = 0$$

$$n(n-10) = 0$$

$$\cancel{n=0} \quad n = 10$$

½ mark for substitution into equation

1 mark for factorial expansion

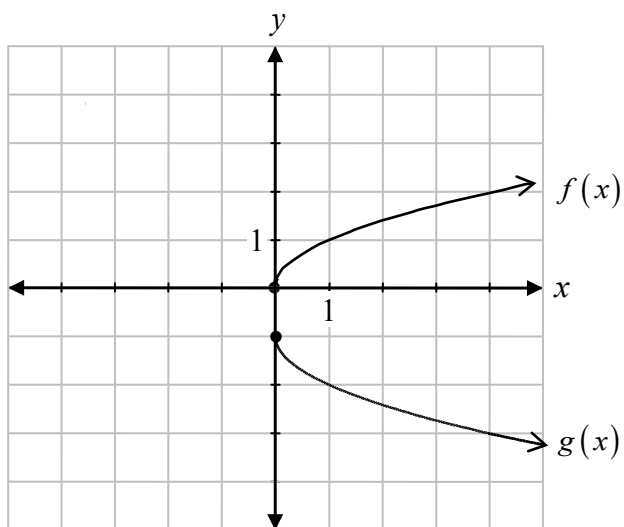
½ mark for simplification of factorials

½ mark for rejecting the extraneous root

½ mark for the value of n

3 marks

Describe the transformations applied to the graph of $f(x)$ to obtain the graph of $g(x)$.

**Solution**

A reflection over the x -axis then a vertical translation of one unit down.

or

A vertical translation of one unit up then a reflection over the x -axis.

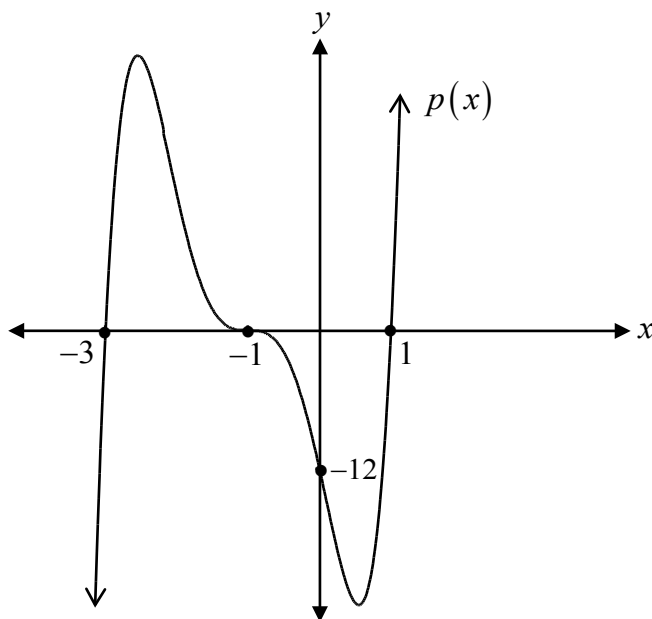
1 mark for reflection over the x -axis
1 mark for vertical translation

2 marks

Note(s):

- Award a maximum of 1 mark if correct transformations are given in the incorrect order.

Determine, algebraically, the value of the leading coefficient of the graph of the polynomial function, $p(x)$.

**Solution**

$$p(x) = a(x+3)(x+1)^3(x-1)$$

$$-12 = a(3)(1)^3(-1)$$

$$-12 = -3a$$

$$a = 4$$

$\frac{1}{2}$ mark for factors of $p(x)$

$\frac{1}{2}$ mark for odd multiplicity for $(x+1)$ greater than 1

$\frac{1}{2}$ mark for substituting $p(0) = -12$

$\frac{1}{2}$ mark for value of a

2 marks

Question 11

P1

Frank, Liam, Chan, and Thao are going to a movie.

Determine the number of ways they can sit in a row of four chairs, if Frank and Chan must sit beside each other.

Solution

$3!2!$

$\frac{1}{2}$ mark for $3!$

$\frac{1}{2}$ mark for $2!$ as a product of factorials

12

1 mark

Determine, algebraically, if $f(x) = \frac{1}{x+5}$ and $g(x) = \frac{1}{x-5}$ are inverses of each other.

Justify your answer.

Solution

Method 1

Let $f(x) = y$

$$y = \frac{1}{x+5}$$

$$x = \frac{1}{y+5}$$

1 mark for switching x and y values

$$y+5 = \frac{1}{x}$$

$$y = \frac{1}{x} - 5$$

$\frac{1}{2}$ mark for solving for y

$$f^{-1}(x) = \frac{1}{x} - 5$$

$$\therefore f^{-1}(x) \neq g(x)$$

$\frac{1}{2}$ mark for justification

2 marks

Method 2

$$f(g(x)) = \frac{1}{\left(\frac{1}{x-5}\right) + 5}$$

1 mark for $f(g(x))$ or $g(f(x))$

$$= \frac{1}{\frac{1+5x-25}{x-5}}$$

$$= \frac{1}{\frac{5x-24}{x-5}}$$

$$= \frac{x-5}{5x-24}$$

$\frac{1}{2}$ mark for simplification

$$g(f(x)) = \frac{1}{\left(\frac{1}{x+5}\right) - 5}$$

$$= \frac{1}{\frac{1-5x-25}{x+5}}$$

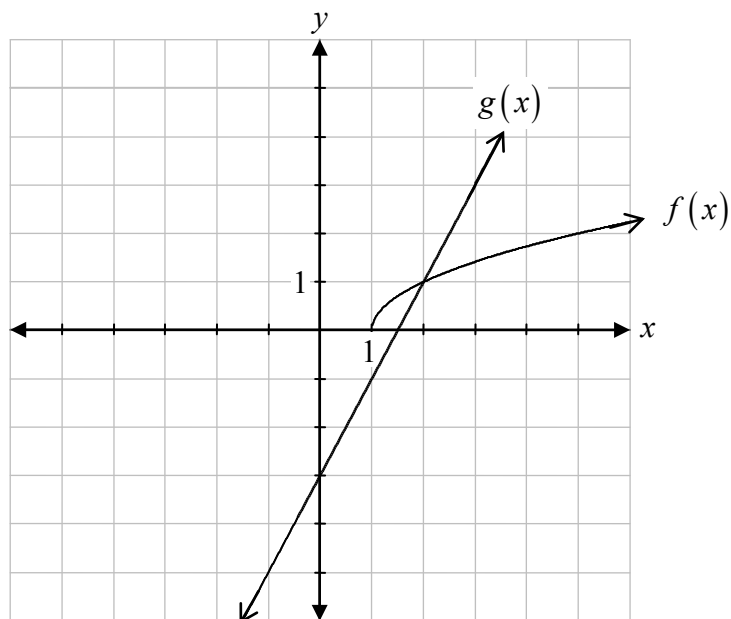
$$= \frac{1}{\frac{-5x-24}{x+5}}$$

$$= \frac{x+5}{-5x-24}$$

$\therefore$ They are not inverses because $f(g(x)) \neq x$ or $g(f(x)) \neq x$. $\frac{1}{2}$ mark for justification

2 marks

Using the graphs of $y = f(x)$ and $y = g(x)$, solve $f(x) = g(x)$.

**Solution**

$$x = 2$$

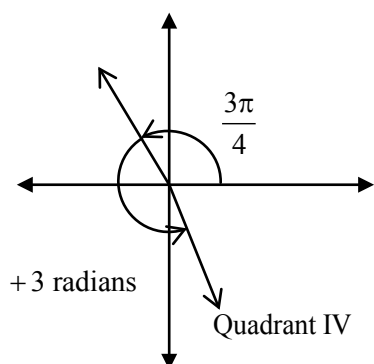
1 mark

An angle in standard position measures $\frac{3\pi}{4}$.

Determine in which quadrant the terminal arm of this angle is located after a rotation of 3 radians.

Justify your answer.

Solution



or

The angle $\frac{3\pi}{4}$ terminates in quadrant II. A rotation of 3 radians is almost a half rotation; therefore, the terminal arm is located in quadrant IV.

1 mark for justification

1 mark

Prove the following identity for all permissible values of θ .

$$\frac{\sin 2\theta}{1 - \cos 2\theta} = \cot \theta$$

Solution

Left-Hand Side	Right-Hand Side
$\frac{\sin 2\theta}{1 - \cos 2\theta}$ $\frac{2 \sin \theta \cos \theta}{1 - (1 - 2 \sin^2 \theta)}$ $\frac{2 \sin \theta \cos \theta}{2 \sin^2 \theta}$ $\frac{\cos \theta}{\sin \theta}$ $\cot \theta$	$\cot \theta$ 1 mark for correct substitution of identities 1 mark for appropriate algebraic strategies 1 mark for logical process to prove the identity 3 marks

If the range of $y = f(x)$ is $-3 \leq y \leq 6$, determine the range of $y = 2f(3x)$.

Solution

$[-6, 12]$

or

$-6 \leq y \leq 12$

1 mark

Maurice incorrectly solved the equation, $\sin \theta + 1 = 0$, over the interval $[0^\circ, 360^\circ]$.

$$\begin{aligned}\sin \theta + 1 &= 0 \\ \sin \theta &= -1 \\ \sin \theta &= 270^\circ\end{aligned}$$

Describe his error.

Solution

Maurice should have written that θ is equal to 270° not $\sin \theta = 270^\circ$.

1 mark

Question 18

R2

If $P(3, 5)$ is a point on the graph of $y = f(x)$, identify the corresponding point on the graph of $y = f(x - 1) + 7$.

a) $(2, 12)$

b) $(4, -2)$

c) $(2, -2)$

d) $(4, 12)$

Question 19

R5

Identify how the graph of $y = 3^x$ is transformed to the graph of $y = 3^{-x}$.

a) reflected over the x -axis

b) reflected over the y -axis

c) reflected over both the x -axis and the y -axis

d) reflected over the line $y = x$

Question 20

R7

Identify the equation $\log_a b = c$ in exponential form.

a) $b^c = a$

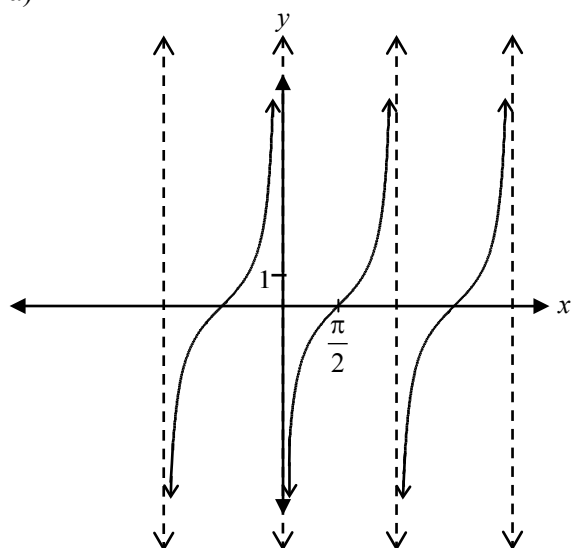
b) $a^c = b$

c) $a^b = c$

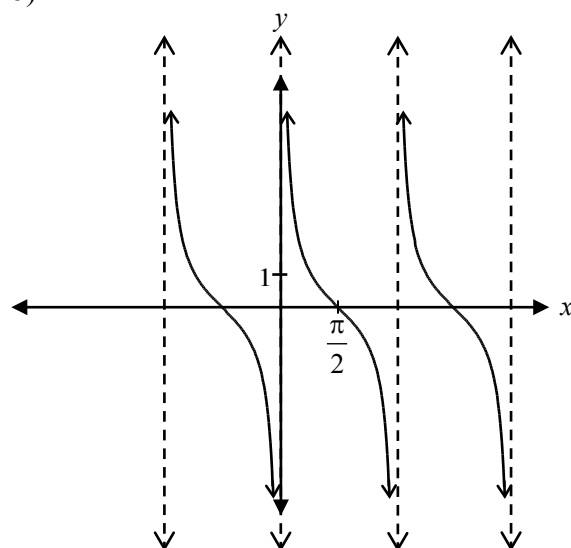
d) $c^a = b$

Identify the graph of $y = \tan x$.

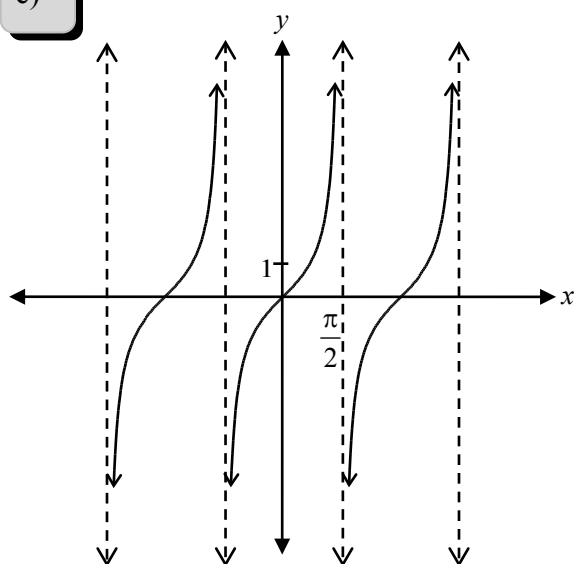
a)



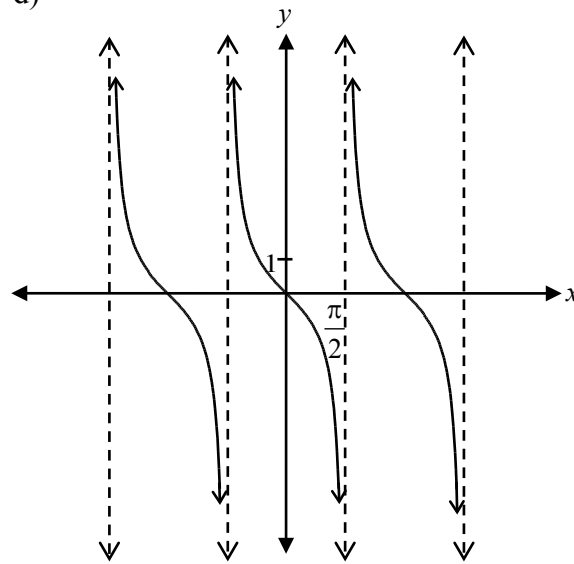
b)



c)

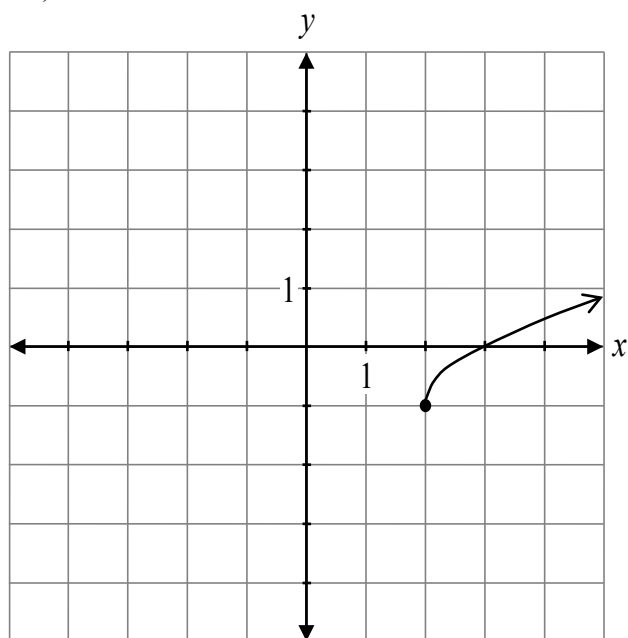


d)

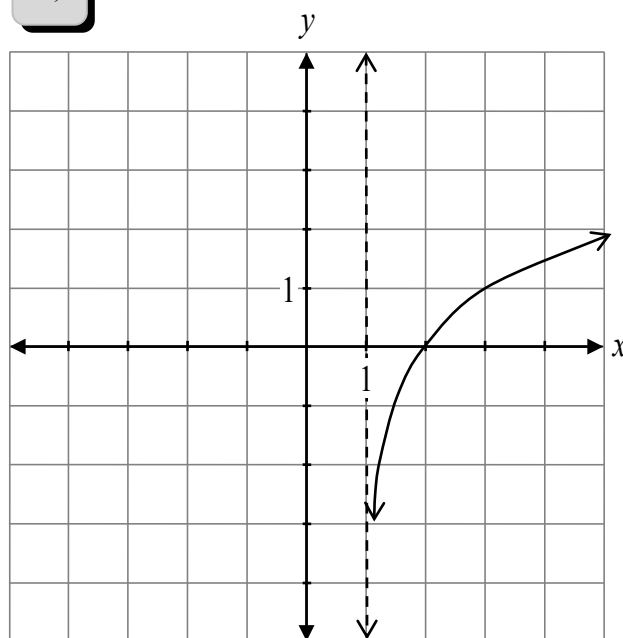


Identify which of the following graphs represents a logarithmic function.

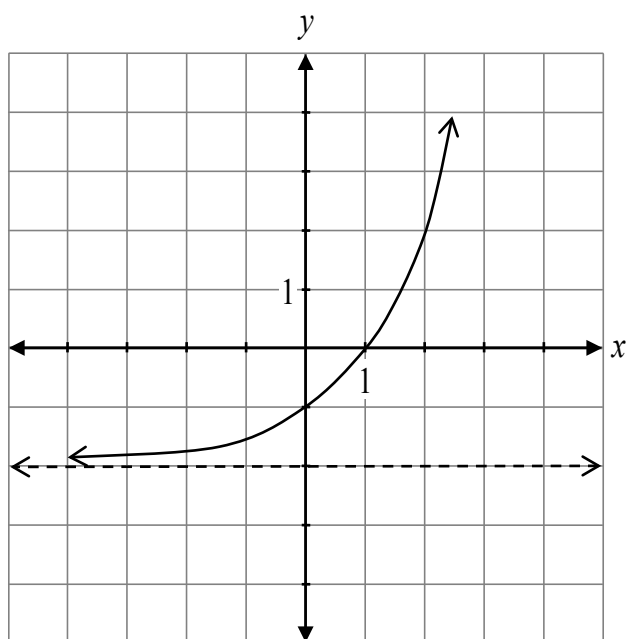
a)



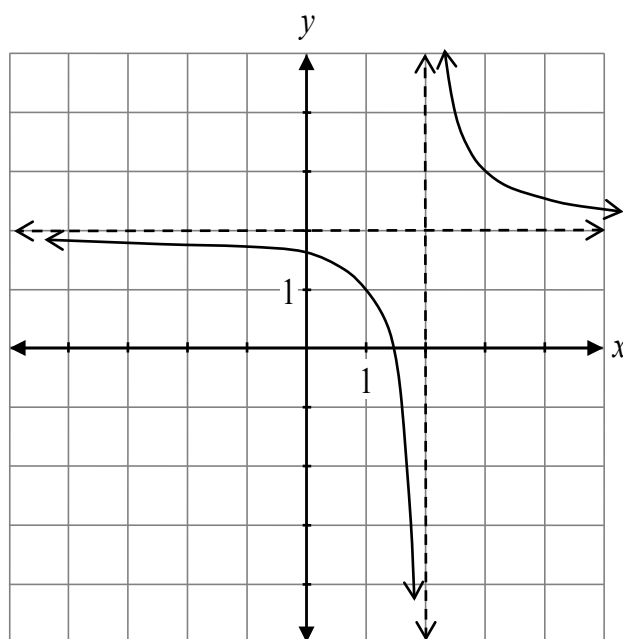
b)



c)



d)



Question 23

R12

If the volume of a box is represented by $V(x) = (x + 4)(x + 2)(x - 1)$, identify a possible value of x .

a) -4

b) -1

c) 1

d) 4

Question 24

T1

Identify a coterminal angle for $\theta = -\frac{\pi}{3}$.

a) $\frac{\pi}{3}$

b) $\frac{4\pi}{3}$

c) $\frac{7\pi}{3}$

d) $\frac{11\pi}{3}$

Question 25

P3

Identify the value of n in the equation ${}_nC_3 = {}_nC_6$.

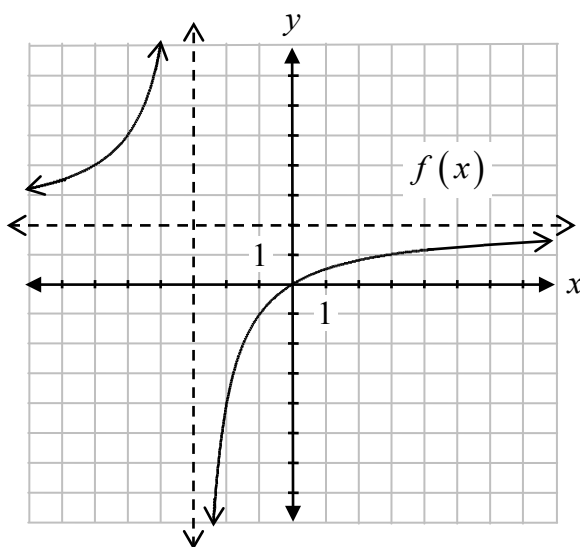
a) 3

b) 6

c) 9

d) 18

Identify the equation of the function, $f(x)$, for the following graph.



a) $f(x) = \frac{2x}{x+3}$

b) $f(x) = \frac{2}{x+3}$

c) $f(x) = \frac{2x^2}{x(x+3)}$

d) $f(x) = \frac{3x^2}{x(x+2)}$

Match the following radical functions with their graphs.

Solution

Place the appropriate letter in this column.

$$f(x) = 2\sqrt{-(x+3)} \quad \underline{\text{B}}$$

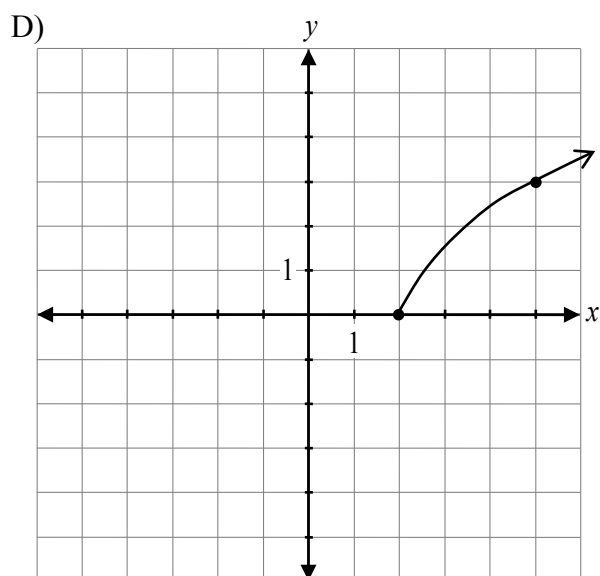
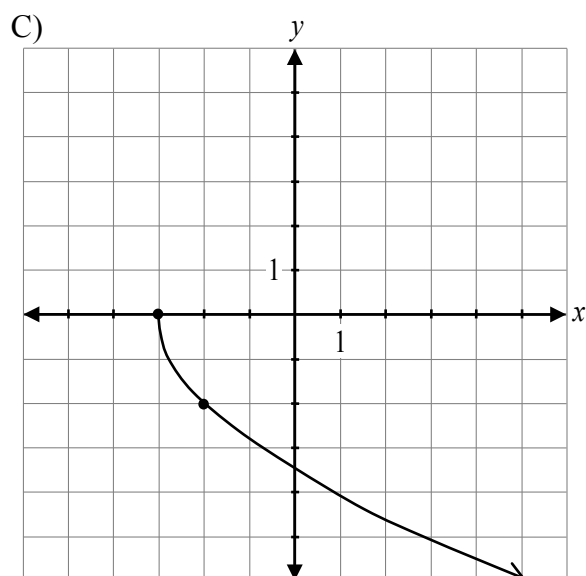
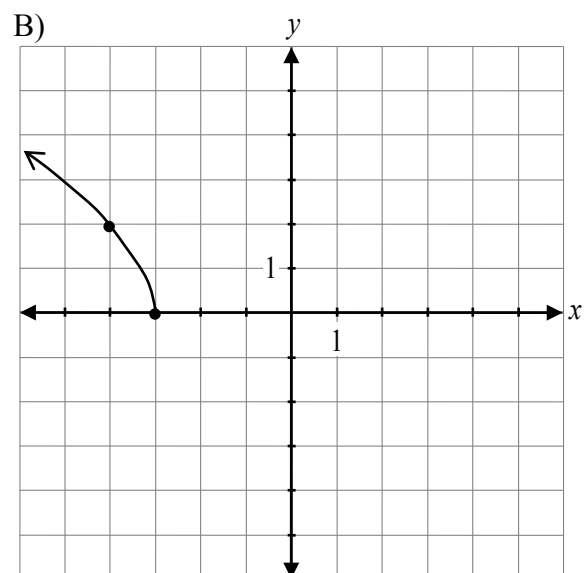
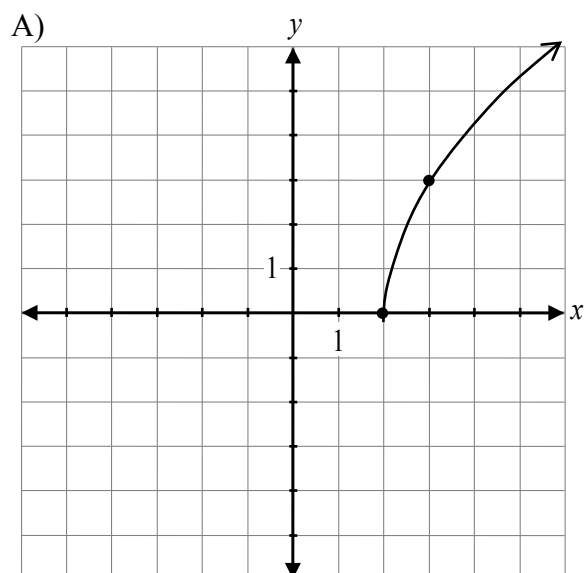
$$g(x) = -2\sqrt{(x+3)} \quad \underline{\text{C}}$$

$$h(x) = 3\sqrt{(x-2)} \quad \underline{\text{A}}$$

$$k(x) = \sqrt{3(x-2)} \quad \underline{\text{D}}$$

$\frac{1}{2}$ mark for each correct answer

2 marks



Express $p(x) = x^3 - 2x^2 - 4x + 8$ as a product of factors.

Solution

$$\begin{aligned} p(2) &= 2^3 - 2(2)^2 - 4(2) + 8 \\ &= 8 - 8 - 8 + 8 \\ &= 0 \end{aligned}$$

1 mark for identifying one possible value of x

$$\begin{array}{r|rrrrr} 2 & 1 & -2 & -4 & 8 \\ & \downarrow & 2 & 0 & -8 \\ \hline & 1 & 0 & -4 & 0 \end{array}$$

1 mark for synthetic division (or any equivalent strategy)

$$p(x) = (x - 2)(x^2 - 4)$$

1 mark for product of factors

or

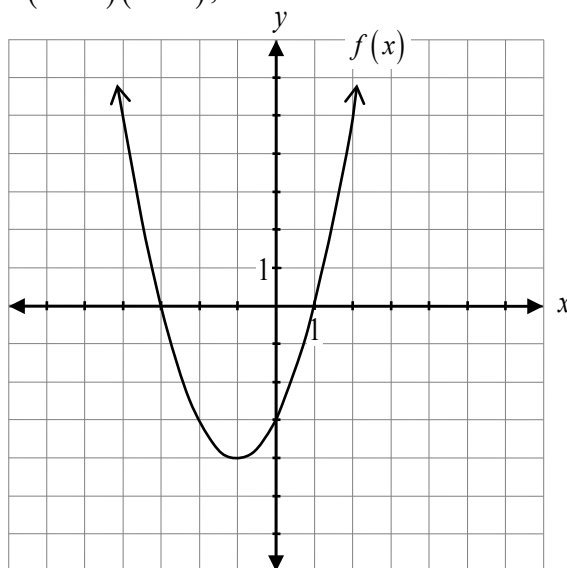
$$p(x) = (x - 2)(x - 2)(x + 2)$$

or

$$p(x) = (x - 2)^2(x + 2)$$

3 marks

Given the graph of $f(x) = (x+3)(x-1)$,

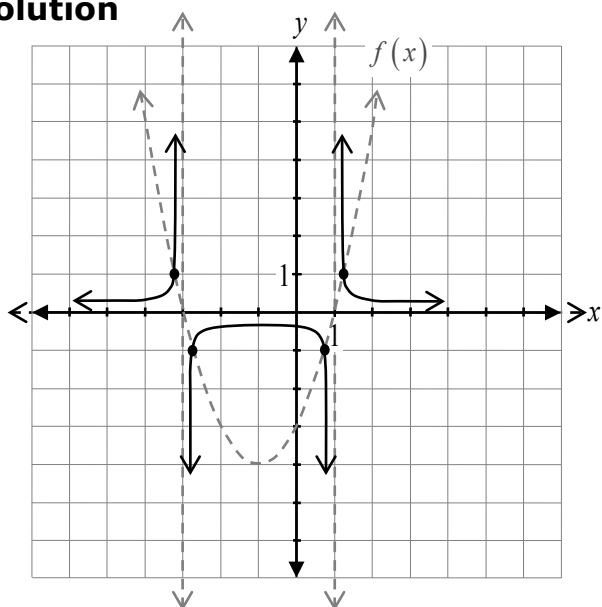


a) sketch the graph of $g(x) = \frac{1}{f(x)}$.

b) describe how to sketch the graph of $h(x) = |f(x)|$.

Solution

a)



1 mark for asymptotic behaviour at $x = 1$ and $x = -3$

$\frac{1}{2}$ mark for asymptotic behaviour at $y = 0$

$\frac{1}{2}$ mark for graph left of $x = -3$

$\frac{1}{2}$ mark for graph between $x = -3$ and $x = 1$

$\frac{1}{2}$ mark for graph right of $x = 1$

3 marks

b) Change all negative y-values to positive values. The positive y-values do not change.

1 mark

Note(s):

- Award 1 mark if only vertical asymptotes at $x = 1$ and $x = -3$ are drawn.

Describe how the value of m in the equation $y = \log_3(x - m)$, $m \in \mathbb{R}$, affects the asymptote on the graph of $y = \log_3 x$.

Solution

The vertical asymptote is translated horizontally left or right m units from the y -axis.

1 mark

Solve algebraically.

$$25^x = \left(\frac{1}{5}\right)^{-3x+1}$$

Solution

$$\left(5^2\right)^x = \left(5^{-1}\right)^{-3x+1}$$

$$5^{2x} = 5^{3x-1}$$

$$2x = 3x - 1$$

$$x = 1$$

1 mark for changing to a common base

½ mark for exponent law

½ mark for equating exponents

2 marks

Solve $\cos 2\theta = 0$, where $\theta \in \mathbb{R}$.

Solution

Method 1

$$1 - 2\sin^2 \theta = 0$$

1 mark for double-angle identity

$$\sin^2 \theta = \frac{1}{2}$$

$$\sin \theta = \frac{\sqrt{2}}{2}$$

$$\sin \theta = \frac{-\sqrt{2}}{2}$$

1 mark for solving for $\sin \theta$

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}$$

$$\theta = \frac{5\pi}{4}, \frac{7\pi}{4}$$

1 mark for values of θ
(½ mark for each branch)

$$\left. \begin{aligned} \theta &= \frac{\pi}{4} + \pi k \\ &= \frac{3\pi}{4} + \pi k \end{aligned} \right\} \text{where } k \in \mathbb{Z} \text{ or } \theta = \frac{\pi}{4} + \frac{\pi k}{2}, \text{ where } k \in \mathbb{Z}$$

1 mark for general solution

4 marks

Method 2

$$2\cos^2 \theta - 1 = 0$$

1 mark for double-angle identity

$$\cos^2 \theta = \frac{1}{2}$$

$$\cos \theta = \frac{\sqrt{2}}{2}$$

$$\cos \theta = \frac{-\sqrt{2}}{2}$$

1 mark for solving for $\cos \theta$

$$\theta = \frac{\pi}{4}, \frac{7\pi}{4}$$

$$\theta = \frac{3\pi}{4}, \frac{5\pi}{4}$$

1 mark for values of θ
(½ mark for each branch)

$$\left. \begin{aligned} \theta &= \frac{\pi}{4} + \pi k \\ &= \frac{3\pi}{4} + \pi k \end{aligned} \right\} \text{where } k \in \mathbb{Z} \text{ or } \theta = \frac{\pi}{4} + \frac{\pi k}{2}, \text{ where } k \in \mathbb{Z}$$

1 mark for general solution

4 marks

Note(s):

- Deduct a maximum of 1 mark if student omits second branch when taking the square root.

Method 3

$$\cos 2\theta = 0$$

$$2\theta = \frac{\pi}{2}, \frac{3\pi}{2}$$

2 marks for values of 2θ (1 mark for each value)

$$\left. \begin{array}{l} 2\theta = \frac{\pi}{2} + 2\pi k \\ 2\theta = \frac{3\pi}{2} + 2\pi k \end{array} \right\} \text{where } k \in \mathbb{Z}$$

1 mark for general solution

$$\theta = \frac{\pi}{4} + \pi k$$

1 mark for values of θ

$$\theta = \frac{3\pi}{4} + \pi k$$

4 marks

Describe a difference between the graphs of $y = f(x)$ and $y = g(x)$.

$$f(x) = -2(x+1)^2(x+3)$$

$$g(x) = 2(x+1)^2(x+3)$$

Solution

The end behaviour is different, because the graph of $f(x)$ falls to the right and the graph of $g(x)$ rises to the right.

or

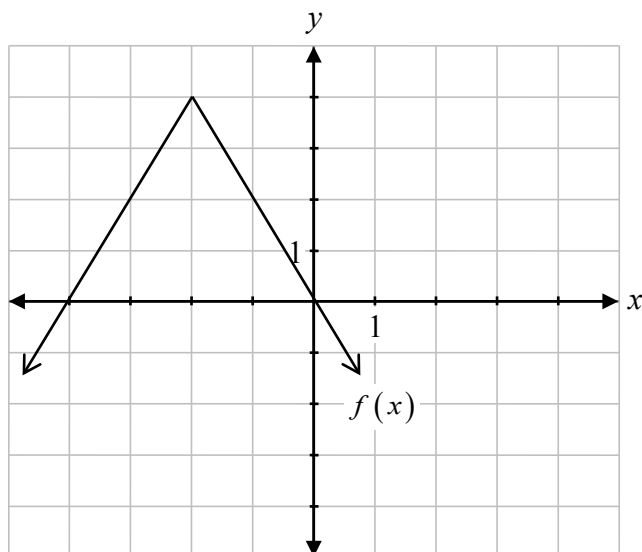
The y -intercept of $f(x)$ is negative while the y -intercept of $g(x)$ is positive.

or

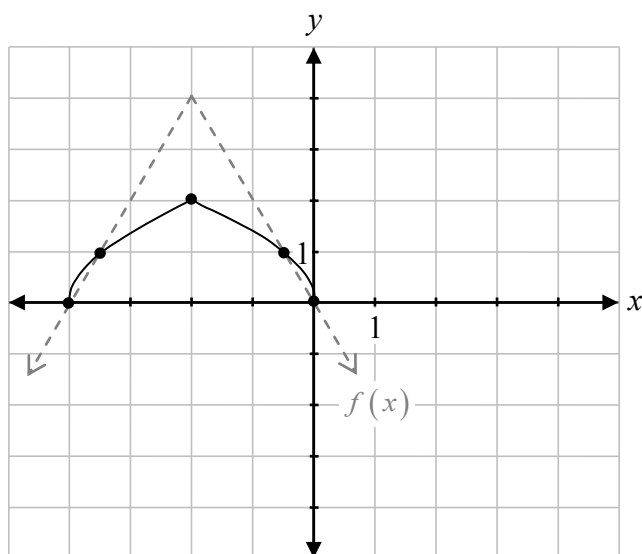
One graph is a reflection over the x -axis of the other graph.

1 mark

Given the graph of $y = f(x)$, sketch the graph of $\sqrt{f(x)}$.



Solution



1 mark for restricted domain

$\frac{1}{2}$ mark for shape between both invariant points,
 $\{0 \leq y \leq 1\}$

$\frac{1}{2}$ mark for shape above invariant points, $\{1 \leq y \leq 2\}$

2 marks

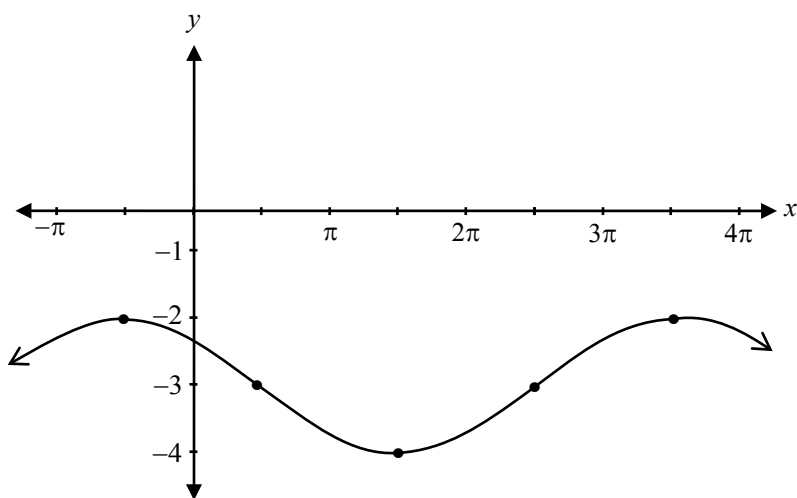
Describe the relationship between the zeros of the function $f(x) = (2x - 1)(x + 3)^2$, the roots of the equation $(2x - 1)(x + 3)^2 = 0$, and the x -intercepts of the graph of $y = f(x)$.

Solution

The zeros, roots, and x -intercepts all have the same values.

1 mark

Sketch a graph of at least one period of the function $f(x) = \cos\left[\frac{1}{2}\left(x + \frac{\pi}{2}\right)\right] - 3$.

Solution

1 mark for period
1 mark for horizontal translation
1 mark for vertical translation

3 marks

Verify that $\theta = \frac{4\pi}{3}$ is a solution of the equation $4 \cos^2 \theta - 1 = 0$.

Solution

$$\text{Left-hand side} = 4 \cos^2 \left(\frac{4\pi}{3} \right) - 1$$

$$= 4 \left(\frac{-1}{2} \right)^2 - 1$$

½ mark for value of $\cos \left(\frac{4\pi}{3} \right)$

$$= 4 \left(\frac{1}{4} \right) - 1$$

$$= 0$$

½ mark for verification

$$= \text{Right-hand side}$$

1 mark

Describe how to determine the equation of the horizontal asymptote of a rational function when the degree of the polynomial in the numerator and the degree of the polynomial in the denominator are equal.

Solution

The horizontal asymptote is $y = \frac{a}{b}$ where a is the leading coefficient in the numerator and b is the leading coefficient in the denominator.

or

Use polynomial division to divide the numerator by the denominator. The equation of the horizontal asymptote will have the same value as the quotient.

1 mark

Evaluate.

$$\frac{\cot\left(-\frac{5\pi}{6}\right)}{\sin\left(\frac{17\pi}{3}\right)}$$

Solution

$$\frac{\sqrt{3}}{-\frac{\sqrt{3}}{2}}$$

$$(\sqrt{3})\left(-\frac{2}{\sqrt{3}}\right)$$

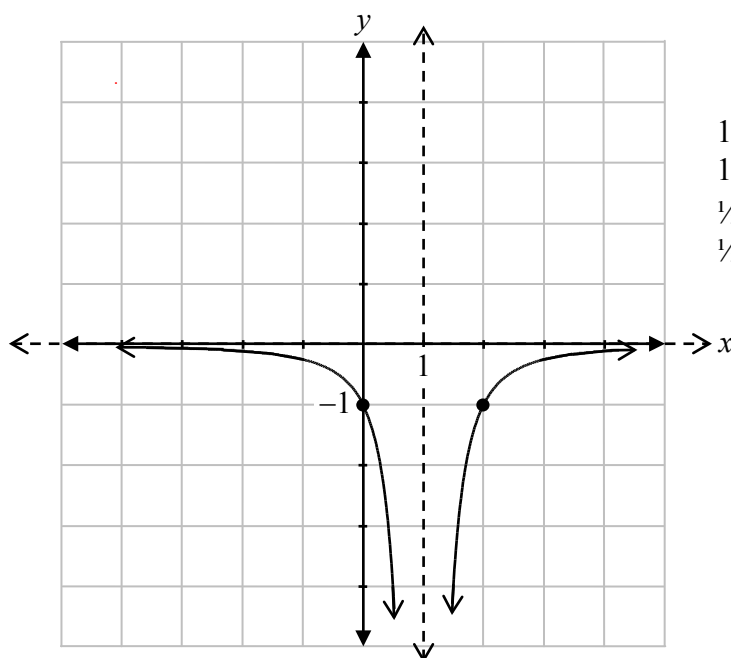
$$-2$$

1 mark for $\cot\left(-\frac{5\pi}{6}\right)$ ($\frac{1}{2}$ mark for value, $\frac{1}{2}$ mark for quadrant)

1 mark for $\sin\left(\frac{17\pi}{3}\right)$ ($\frac{1}{2}$ mark for value, $\frac{1}{2}$ mark for quadrant)

2 marks

Sketch the graph of the function $f(x) = \frac{-1}{(x-1)^2}$ and determine the range.

Solution

1 mark for asymptotic behaviour at $x = 1$
1 mark for asymptotic behaviour at $y = 0$
 $\frac{1}{2}$ mark for graph left of $x = 1$
 $\frac{1}{2}$ mark for graph right of $x = 1$

Range: $\{y \in \mathbb{R} \mid y < 0\}$

or

Range: $(-\infty, 0)$

1 mark for range (consistent with graph)

4 marks

Given $f(x) = \sqrt{x-2}$ and $g(x) = x^2 + 1$,

- a) determine $g(f(x))$.
- b) explain why the domain of $g(f(x))$ is restricted.

Solution

a) $g(f(x)) = (\sqrt{x-2})^2 + 1$ 1 mark for composition
 $= x - 2 + 1, x \geq 2$
 $= x - 1, x \geq 2$

1 mark

- b) The domain of $g(f(x))$ must be restricted because the domain of $f(x)$ is restricted.

or

The value of the radicand must be positive.

1 mark

Solve algebraically.

$$2\log_a 3 + \log_a 4 = 2, \text{ where } a > 0$$

Solution

$$\log_a (3^2 \cdot 4) = 2$$

$$\log_a 36 = 2$$

$$a^2 = 36$$

$$a = 6$$

1 mark for power law

1 mark for product law

1 mark for exponential form

3 marks

Solve $\sec \theta + 2 = 0$ over the interval $[0, 2\pi]$.

Solution

$$\sec \theta + 2 = 0$$

$$\sec \theta = -2$$

$$\cos \theta = -\frac{1}{2}$$

$$\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$$

1 mark for reciprocal

1 mark for values of θ ($\frac{1}{2}$ mark for each value)

2 marks

Determine the x -intercept of the graph of $f(x) = e^x - 1$.

Solution**Method 1**

$$0 = e^x - 1$$

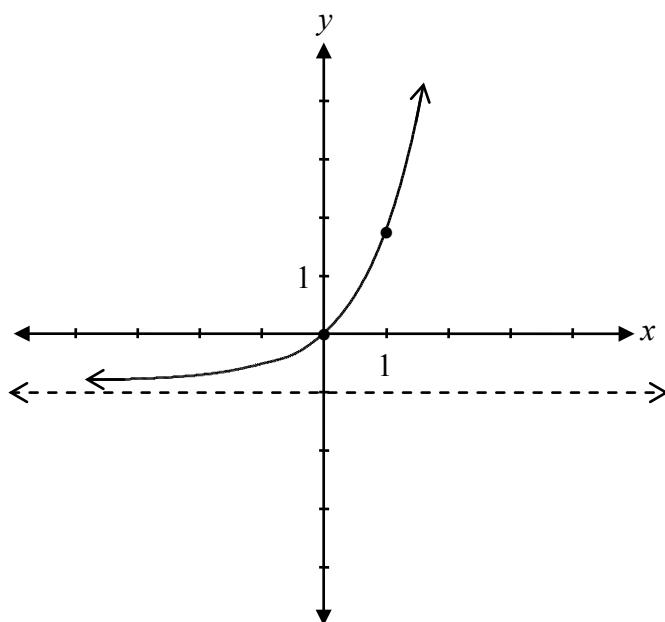
$\frac{1}{2}$ mark for substitution

$$1 = e^x$$

$$x = 0$$

$\frac{1}{2}$ mark for solving for x

1 mark

Method 2

$$\therefore x = 0$$

1 mark

Question 45

P4

Given the 5th row of Pascal's triangle, determine the values of the next row.

1 4 6 4 1

Solution

1 5 10 10 5 1

1 mark

Evaluate.

$$\log_2 80 - \log_2 10$$

Solution

$$\log_2 \left(\frac{80}{10} \right)$$

1 mark for quotient law

$$\log_2 8$$

$$3$$

1 mark for evaluating the logarithm

2 marks

State the amplitude of $f(x) = -2 \sin(x - \pi) - 1$.

Solution

2

1 mark

Determine the exact value of $\cos 15^\circ$.

Solution

$$\cos(15^\circ) = \cos(60^\circ - 45^\circ)$$

$$= \cos 60^\circ \cos 45^\circ + \sin 60^\circ \sin 45^\circ$$

1 mark for substitution into correct identity

$$= \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right)$$

2 marks for values ($\frac{1}{2}$ mark for each exact value)

$$= \frac{\sqrt{2}}{4} + \frac{\sqrt{6}}{4}$$

3 marks

$$= \frac{\sqrt{2} + \sqrt{6}}{4}$$

Note(s):

- Other combinations are possible.

Question 49

R1

Given $f(x) = x^2 + 5x + 6$, $g(x) = x + 3$, and $h(x) = f(x) - g(x)$,

a) determine $h(x)$.

b) sketch the graph of $y = h(x)$.

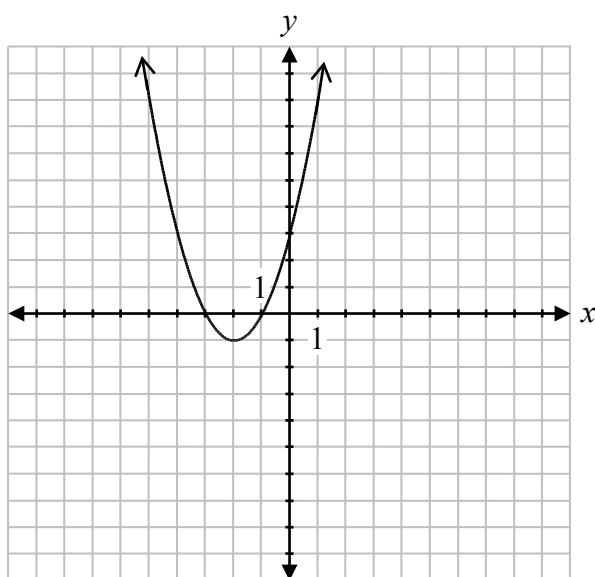
Solution

a)

$$\begin{aligned} f(x) - g(x) &= x^2 + 5x + 6 - (x + 3) \\ &= x^2 + 4x + 3 \end{aligned}$$

1 mark

b)



1 mark for graph consistent with a)

1 mark